

The Birth of the Universe

The prevailing cosmological model suggests that the universe began approximately 13.8 billion years ago in an event known as the Big Bang. Rather than an explosion into empty space, it was an expansion of space itself.

As the universe expanded it cooled, allowing fundamental particles to combine into atoms. Gravity gradually gathered matter into stars and galaxies.

Modern observatories study the cosmic microwave background, galaxy distributions, and gravitational waves to understand the earliest moments of cosmic history.